# Hierarchical Federated Learning With Social Context Clustering-Based Participant Selection for Internet of Medical Things Applications

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***Abstract*— The proliferation in embedded and communication technologies made the concept of the Internet of Medical Things (IoMT) a reality. Individuals' physical and physiological status can be constantly monitored, and numerous data can be collected through wearable and mobile devices. However, the silo of individual data brings limitations to existing machine learning approaches to correctly identify a user's health status. Distributed machine learning paradigms, such as federated learning, offer a potential solution for privacy-preserving knowledge sharing without sending raw personal data. However, federated learning is vulnerable to harmful participants that can degrade the overall model quality by sharing low-quality data. Therefore, it is critical to select suitable participants to ensure the accuracy and efficiency of federated learning. In this article, a unique clustering-based approach is proposed to use social context data for participant selection. Different edge participant groups will be established, and group-specific federated learning will be performed. The models of various edge groups will be further aggregated to strengthen the robustness of the global model. The experimental results demonstrated that through participant selection, clustering-based hierarchical federated learning can achieve better results with less participants in two different IoMT applications for ECG and human motion monitoring. This shows the efficacy of the proposed method in improving federated learning performance and efficiency in various IoMT applications.**

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## I. Introduction

THE widespread of interconnected embedded, wearable, and mobile devices has enabled the collection of large amounts of social, physical, and physiological data of individual users. These data can offer invaluable insight to one's activity and health status. However, these data are largely isolated due to privacy reasons and are not sufficient to train good models for individuals with existing machine learning or deep learning-based approaches.

Distributed machine learning paradigms, such as federated learning, provide a promising solution to overcome the issue of data scarcity of individual users. Federated learning allows participants to share their knowledge safely without transmitting the collected raw data [1], [2]. As such, a global model can be trained with all participants' knowledge (i.e., their model parameters) in a more privacy-preserving manner. The privacy-preserving and knowledge sharing features make federated learning a promising solution for several Internet of Medical Things (IoMT)-related applications [3], such as privacy protection for smart healthcare systems [4] and edge computing-based clinical decision systems [5]. However, similar to other distributed machine learning approaches, the quality of aggregated model depends on the quality of the shared knowledge. Therefore, federated learning is vulnerable to low-quality participants [6].

With or without malicious intent, participants may have very different features due to different physical and social contexts (e.g., age, gender, ethnicity group, weight, and height). Such a difference may lead to very different models. For example, a behavior model of a 20-year-old young person may be very different from a mid-age person's (e.g., 50 years old) behavior model. Knowledge sharing and aggregating parameters of such models are likely to degrade the global model performance. Therefore, this study focuses on the approach of selecting a good set of participants to ensure the efficiency and accuracy of the conducted federated learning.

In addition to the data collected through IoMT, modern social media networks are also able to collect a wide range of unstructured data such as age, gender, ethnicity group,

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physical/mental status, and lifestyle. The collected context information can offer good features to separate people into groups with higher similarity, where federated learning can be conducted with better accuracy and efficiency. Such social context data can offer valuable features for selecting participants in the process of federated learning.

To address the issues of data privacy, data scarcity, and low-quality (or even malicious) participants, a two-layer hierarchical federated learning framework with clustering-based participant selection is proposed in this study. The clustering technique will form multiple user groups with similar social context, where individual groups conduct their own federated learning at the edge layer. To further strengthen the robustness of federated learning, a hierarchical approach is adopted to design a second layer of aggregation, where models produced by different edge groups are aggregated at the cloud layer. A unique aggregation function is designed based on the model quality, which is estimated by the cluster quality estimated through the silhouette coefficient. This again ensures that models with very different contexts do not interfere with each other too much while improving the generality and robustness of the trained global model.

The contributions of this work are summarized as follows.

1) A unique hierarchical federated learning framework is proposed to integrate unstructured social context data collected with an end-edge-cloud infrastructure to achieve better federated learning accuracy and efficiency.
2) A Gaussian mixture model (GMM) clustering-based participant selection approach is proposed to group participants with similar social context to improve federated learning accuracy and efficiency by reducing the potential impact of low quality, dissimilar, or even adverse participants data.
3) A unique cloud-level aggregation function based on the cluster quality using silhouette coefficient, which further enhances the robustness and generality of the federated learning models.

The rest of this article is organized as follows. Section II provides an overview of the state-of-the-art works on the IoMT and federated learning targeting modern Internet of Things (IoT) or IoMT infrastructures. Section III describes the IoMT motivating scenario and the overview of the proposed clustered hierarchical federated learning framework. The detailed clustered hierarchical federated learning methodology is explained in Section IV, followed by the experimental results presented in Section V. Section VI provides the concluding remarks and our future research perspectives.

## II. Related Work

In this section, the state of the art of federated learning research applied in modern IoT (or end-edge-cloud) infrastructures, and research targeting IoMT applications is summarized.

### A. Federated Learning in End-Edge-Cloud Computing

Federated learning, known as "machine learning at the edge," is first proposed in 2016 [7]. It is a new paradigm of distributed machine learning, allowing local or edge servers to collaboratively train, share, and update one learning model without sharing their private data. This brings forward new challenges in privacy, communication, latency, connectivity, heterogeneity, and so on [8]. Ghimire and Rawat [9] compared federated learning with centralized learning and distributed on-site learning and surveyed successful adoption of federated models in terms of their performance in privacy and security issues for IoT systems. Ferrag et al. [10] studied the performance of federated learning considering three real IoT datasets and provided a review on federated learning based on the experimental analysis in different types of IoT applications, such as Industrial IoT, Internet of Vehicles, and Internet of Healthcare Things. Gu et al. [11] designed a so-called synchronous federated stochastic gradient descent algorithm for multiparty collaborative learning. They further proposed two variance reduction variants for federated learning using vertically partitioned data and analyzed their convergence rates in terms of computational complex and communication cost. Mills et al. [12] focused on the multitask federated learning scheme and integrated a nonfederated batch normalization layer into the federated deep neural network, aiming to realize personalized model training based on users' local data in edge computing environments. Cao et al. [13] incorporated generative adversarial network (GAN) into federated learning to facilitate personalized training in each client, and thus, the client could design its own model and participate in federated learning independently without sharing any model structure or information with other clients. To alleviate communications between client and server, Chen et al. [14] presented an asynchronous learning scheme, in which a temporally weighted aggregation algorithm was employed when training the local models, thus could reduce the number of parameters, and update them in the shallow and deep layers with fewer frequencies. Messaoud et al. [15] introduced a federated model with deep reinforcement learning scheme for resource allocation and management in industrial IoT networks, in which a multiagent deep Q-learning scheme was developed to maximize the self-quality of service (QoS) requirement, while a deep federated learning scheme was developed to build the global model from multiagent self-models and finally find the optimal action decision. Su et al. [16] focused on the concept of artificial intelligence (AI) of Things and utilized a federated learning scheme to deal with the communication-efficient energy data sharing for smart grids in edge-cloud computing. They then addressed a two-layer reinforcement learning scheme with an evaluation mechanism for local data, to trade off the accuracy loss and expenditure in federated learning. Jiao et al. [17] constructed an auction-based market model for wireless federated learning, in which they employed deep reinforcement learning and graph neural networks to implement an automated strategy-proof mechanism for social welfare improvement in a federated learning platform. Lin et al. [18] built a semi-decentralized learning architecture for federated learning in device-to-device communications, in which an adaptive control algorithm was developed to minimize a tradeoff between consumption, delay, and accuracy based on the cooperative local clusters and model aggregations. Sattler et al. [19] proposed so-called clustered federated learning to deal with the multitask learning problem.

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They considered the similarity between clients in terms of their gradient weight updates and divide the client population into several clusters according to the geometric properties of loss surface and joint trainable data distributions as well, in order to learn more specialized models.

### B. Applications on Internet of Medical Things

As an important extension or branch of IoT in e-health, IoMT has exhibited its advantage in collecting, transmitting, and analyzing medical data based on interconnected biomedical sensors, which greatly enables remote diagnosis, control, and monitoring in smart healthcare systems [20], [21], [22]. Philip et al. [23] pointed out that advances in IoT technologies, including optimized network architecture, increased device reliability, and low power consumption, were becoming the important catalysts in providing personalized and high-quality healthcare services, especially for in-home health monitoring systems. Gatouillat et al. [24] summarized the recent contributions on better robustness, reliability, and safety when using medical-grade devices to construct wider scale health networks in IoMT for both patients and healthcare providers. Sun et al. [25] focused on how to deal with the high-quality medical resources in a timely manner and analyze the medical big data in medical applications. In particular, they conducted a survey on the state-of-the-art techniques in edge-cloud computing and AI models that are used in IoMT for high-quality medical services. Considering the security issue in IoMT, Ghubaish et al. [26] reviewed the available security technologies against a variety of attacks and proposed a security framework using a combination of some of these technologies, to enhance the data collection, storage, and transmission in IoMT systems. Indumathi et al. [27] investigated the intersection of blockchain, cloud computing, and IoMT in healthcare and formulated the critical requirements and functions for real-time remote support in patient-centric healthcare. Guo et al. [28] presented a hybrid recognition framework that combined the advantages of deep machine learning and conventional machine learning, which could be used for remote patient diagnosis in IoMT based on the medical image analysis. Su et al. [29] built a valvular heart disease screening system, in which an IoMT module based on a deep learning algorithm was developed to enhance the curve fitting and analysis. Deebak and Al-Turjman [30] proposed a smart service authentication framework based on IoMT, in which both the formal and informal verification methods were considered to cross-examine the common secret session key for better mutual authenticity. Manogaran et al. [31] focused on the uncertainty in the IoMT platform and discussed cognitive data processing for uncertainty analysis in terms of the aggregation and dissemination of wearable sensor data. They argued that the joint learning process of regression learning and conditional sigmoid function analysis could be useful to improve the efficiency of data management in smart medical healthcare systems.

### C. Participant Selection in Federated Learning

While there are obvious benefits for federated learning, it also faces some well-known challenges, such as device and data heterogeneity, network dynamicity, and trustworthiness [32]. Several existing participant selection techniques are developed to address device heterogeneity within the federated learning process using threshold-based or reinforcement learning-based approaches [33], [34]. There are also other techniques proposed to deal with data heterogeneity and non-IID issues [35]. However, very limited work is on clustering-based participant selection. In [36], a clustered sampling approach is proposed to achieve unbiased and low-variance selection. Our proposed method aims to find similar participants through clustering using social context data, which, to the best of our knowledge, is unique and different from existing methods.

## III. Overview of the Target Scenario and Framework Architecture

### A. Target IoMT Scenario

Modern wearable and mobile devices allow healthcare data to be monitored and collected in real time in a remote and distributed fashion. The availability of real-time data brings hope to offer distributed and high-precision healthcare services. However, due to the distributed nature, the collected data are in their silos. The existing centralized machine learning paradigms cannot effectively use those data to provide high-quality healthcare services, primarily due to privacy and security concerns.

Referring to Fig. 1, a typical distributed IoMT scenario is presented, where multiple users are monitored by their own wearable and mobile devices. As mentioned previously, cloud-based centralized learning that aggregates all participants' data can be inefficient and ineffective. On the other hand, data collected from an individual are also insufficient to train an accurate model. Through secure sharing and aggregating knowledge across multiple users, federated learning offers the capability to solve data scarcity faced by many existing IoMT applications.

However, it is challenging to ensure that all the participants are contributing toward the aggregated model in a positive and efficient manner. In reality, participants with different social and/or physical contexts, for example, age, gender, ethnicity group, weight, height, and lifestyle, can lead to very different behavior or healthcare models. Therefore, unselectively combining all participants may not be the best choice and may not result in the most accurate and efficient federated learning process.

In this study, we aim to investigate the typical IoMT scenario shown in Fig. 1 in a modern end-edge-cloud infrastructure, combining social context information and wearable sensor data to achieve more accurate and efficient federated learning. This target scenario will be demonstrated in Section V using two real-world IoMT databases for ECG and human motion monitoring.

### B. Overview of the Proposed Clustered Hierarchical Federated Learning Framework

Based on the target scenario, an overview of the proposed clustered federated learning framework is shown in Fig. 1,

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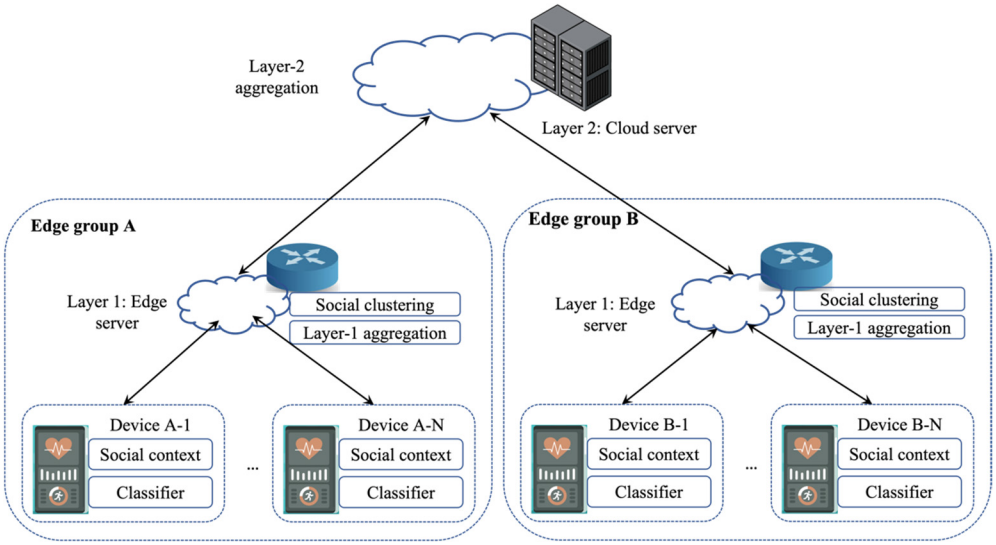



Fig. 1. Hierarchical federated learning with clustering-based participant selection framework for distributed IoMT applications in an end-edge-cloud infrastructure.

where individuals are equipped with various types of ambient or wearable devices that constantly monitor their physical activities and physiological status. In addition, mobile devices (e.g., smartphones) also collect their social context information, such as age, gender, ethnicity group, weight, height, and lifestyle.

The social context information in this case is considered as inputs into the participant selection clustering algorithm. The participant selection algorithm will allocate individuals into a specific learning group (i.e., cluster) based on their similarity in terms of social context. In this work, our hypothesis is that individuals who share similar social context will have a higher likelihood to have similar behavior or healthcare data distributions, and hence, the collected data are more likely to contribute positively to the federated learning model. Therefore, a clustering-based approach is proposed and used to form multiple learning groups, aiming to achieve better federated learning performance. After the group formation, each individual group will conduct its own federated learning at the edge layer. With less amount of participants that share similar social context, federated learning of each individual group is expected to produce more accurate results with better learning efficiency.

Depending on the participant selection algorithm, multiple groups will be formed, and inevitably, some group might be smaller. To further strengthen the robustness of the federated learning models and achieve the benefits of quality knowledge sharing, a second layer of federated learning is conducted by aggregating the models of individual edge groups with a group-specific aggregation strategy, which ensures that model parameters coming from similar social context will be considered more valuable. The second-layer aggregation will be conducted periodically to ensure a more robust and more generalized model.

## IV. Federated Learning With Clustering-Based Participants Selection

Based on the framework overview, this section will provide the detailed descriptions of the proposed methodologies. The fundamental idea of this study is to demonstrate that federated learning can be more efficient by making use of social context information. The first subsection introduces the adopted clustering algorithm for the purpose of participant selection and formation of federated learning edge groups with similar social contexts. Based on the grouping, the proposed hierarchical federated learning will be explained and its algorithm is presented in Section IV-B.

### A. Social Context-Based Participant Selection

Following the target scenario presented in Section III-A, the main challenges of modern IoMT applications are two folds. First, individual's IoMT data are likely to be insufficient for modern deep learning-based approaches and cloud-based centralized learning approaches cannot offer efficient and high-quality services using real-time IoMT data. A federated learning-based approach can nicely address this challenge through secure knowledge sharing. However, the knowledge sharing nature of federated learning brings the second challenge, which is the necessity of ensuring the participant quality. The overall performance of federated learning is highly dependent on the quality of participant models. Therefore, the fundamental idea of our study is to make use of the social context data to allocate participants into separate learning groups that can result in more accurate and efficient federated learning.

Referring to Fig. 1, the participants within an edge group will be clustered according to their social context data, that is, the grouping is done by a clustering algorithm that uses social context as features. As a result, participants with similar social contexts will be allocated into the same group. In this study, the classic distribution-based clustering method, the GMMs, is used to process the social context data and to automatically allocate participants into a specific learning group. The GMM-based clustering algorithm is defined by the following $D$-dimensional Gaussian probability density function

$$p(\mu, \Sigma) = \frac{1}{(2\pi)^{\frac{D}{2}}\sqrt{|\Sigma|}} \exp\left[-\frac{1}{2}(\mu - x)^T \Sigma^{-1}(x - \mu)\right] \quad (1)$$

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where $x$ is the data points, $D$ is the dimension of each data point, $\mu$ is the mean, and $\Sigma$ is the covariance.

Considering the target IoMT scenario, where social context data may be incomplete, the GMM-based clustering approach is picked for its ability to find subgroups within an overall population in an unsupervised manner. This is one of the key advantages of GMM-based clustering, where the label of which group a data point belongs to is not required.

However, one of the potential vulnerabilities of GMM clustering is the likelihood of overfitting, which will be addressed by the hierarchical federated learning approach proposed in Section IV-B, where models from different groups are merged with different weightings. One purpose of hierarchical federated learning is to ensure the robustness of the behavior and healthcare models with a certain level of generality.

### B. Hierarchical Federated Learning

Once the participants are separated into their specific groups, the first layer of federated learning begins within each individual group at the edge layer. The purpose of this layer is to achieve knowledge sharing within a group to address the issue of data scarcity. This layer of federated learning can be achieved at the edge level to enhance the communication and learning efficiency. In this study, typical horizontal federated learning is used to exchange and aggregate knowledge within each participant group.

For each participant, a convolutional neural network (CNN) with two pairs of pooling and convolution layers, followed by a fully connected layer, is used, as defined in (2). The first-layer federated learning aggregation is done by pulling each participant within a group after an iteration of a local CNN training

$$h(x_i, \omega) = \mathrm{FC}(\mathrm{Pool}(\mathrm{Conv}(\mathrm{x_i}, \omega_{\mathrm{Conv}}), \omega_{\mathrm{Pool}}), \omega_{\mathrm{FC}}) \quad (2)$$

where $x_i$ is the data samples of an individual participant $i$; Pool$(*)$, Conv$(*)$, and FC$(*)$ refer to the pooling, convolution, and fully connected layers, respectively; and $\omega_*$ is the parameter of the corresponding layer in the model.

As mentioned in Section IV-A, one of the vulnerabilities of GMM-based clustering is overfitting. The same issue can also happen at the first layer of federated learning. Therefore, the second layer of federated learning is introduced to ensure a balance between achieving a robust and general model while ensuring that irrelevant participant models are not included to prevent degradation of the overall model quality.

In this study, the clustering performance (and hence the quality of selected participants) in the first layer is evaluated using the silhouette coefficient. Each edge group will calculate and transfer the silhouette coefficient of its clustering to the cloud server, where the coefficient will be used to work out the weighting of the second-layer model aggregation.

The second-layer federated learning is achieved using a unique aggregation function based on the Euclidean distance between individual cluster centers. The assumption is that the distance between cluster centers can give an indication to the relevance between two different clusters. The cloud server will find and group clusters that are close to each other (i.e., with similar cluster centers). Within each group of clusters, the first-layer federated learning models will be aggregated according to their quality measured by the silhouette coefficient in (3) and (4). The model that came from a better-quality clustering (i.e., higher silhouette score) will take a higher weighting in the second-layer aggregation process, as given in the following equations:

$$\text{Silhoutte coefficient, } S_i = \frac{D - d}{\max(D, d)} \quad (3)$$

$$\text{Cluster silhoutte score, } \mathrm{CS}_j^k = \sum_i \frac{S_i}{N} \quad (4)$$

$$w_j^k = \mathrm{CS}_j^k \Big/ \sum_k \mathrm{CS}_j^k \quad (5)$$

where $i$ is the data sample index, $N$ is the total number of data samples, $j$ is the cluster index, $k$ is the edge group index, $\mathrm{CS}_j^k$ is the silhouette coefficient of cluster $j$ from edge group $k$, and $w_j^k$ is the weight of cluster $j$ from edge group $k$ in the global model aggregation.

**Algorithm 1** Hierarchical Federated Learning With Clustering-Based Participant Selection Algorithm for IoMT Applications

**Input:** A set of participant clusters/groups $PG = \{pg_i\}_{i=1}^{G}$
A set of participants in each group $P = \{pg_n^g\}_{n=1}^{N}$
**Output:** A set of trained global model $M_i$
1: Initialize model parameter, maximum iterations $T$
2: **for** $t = 1$ to $T$ **do**
3: **for** each group $pg_g \in PG$ **do**
4: **for** each participant $p_n \in P$ **do**
5: Conduct local train for the model $m_n$ on $p_n$ by Eq. (2)
6: Submit the local learning parameter $\omega_i(t)$
7: **end for**
8: Aggregate the first layer local parameters $\omega_i(t)$
9: **end for**
10: **if** $t = mod(T, 10)$ **do**
11: Calculate the sample silhouette coefficient by Eq. (3)
12: Calculate the cluster silhouette coefficient by Eq. (4)
13: Calculate the weight for 2nd-layer aggregation Eq. (5)
14: Broadcast $\omega_i(t+1)$ to the participant group $pg_i$
15: **end for**
16: **return** $M_i$

The overall clustered hierarchical federated learning algorithm is presented in Algorithm 1. Lines 2–9 correspond to the first-layer federated learning conducted on specific clusters. Within each edge group, $j$ clusters will be formed. In every tenth iteration of the first-layer federated learning, the cluster-specific model will be submitted to the second layer to be aggregated with the corresponding clusters from different edge groups, which is shown in lines 10–16.

## V. Experiments

### A. ECG and Human Activity Recognition Datasets for IoMT Scenario Evaluation

While IoMT technologies and applications are more commonly available nowadays, collecting and labeling healthcare data remain a very time-consuming process. In this study, we selected two distinctive IoMT application datasets, namely,

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TABLE I
AGE AND GENDER DISTRIBUTION OF THE MIT-BIH ECG DATASET

DATASET

| | MIT-BIH | |
|---|---|---|
| Age | gender | |
| | Male | Female |
| 50-60 | 5 | 2 |
| 60-70 | 12 | 2 |
| 70-80 | 2 | 6 |
| 80-90 | 3 | 6 |

TABLE II
ANNOTATED CLASSES FOR THE MIT-BIH AND HAR DATASETS

| MIT-BIH Classes | HAR Classes |
|---|---|
| normal_beats = ['N', 'L', 'R', 'e', 'j'] | downstairs = {'dws': 0} |
| sup_beats = ['A', 'a', 'J', 'S'] | jogging = {'jog': 1} |
| ven_beats = ['V', 'E'] | sitting = {'sit': 2} |
| fusion_beats = ['F'] | standing = {'std': 3} |
| unknown_beat = ['/', 'f', 'Q'] | upstairs = {'ups': 4} |
| - | walking = {'wlk': 5} |

MIT-Beth Israel Hospital (BIH) and human activity recognition (HAR), which contain ECG and motion data, respectively, to evaluate the performance and effectiveness of the proposed approach. These two datasets are chosen deliberately to demonstrate the ability of the proposed methods in handling different IoMT applications.

The first dataset is the MIT-BIH database, which was collected for general cardiac-related research [37], [38]. This dataset contains only the gender and age information. The gender and age divisions are summarized in Table I. The database contains 15 different labels and was further categorized into five different classes according to the IEC/AAMI ECG standards, as summarized in Table II. The ECG data are collected from 47 subjects, with 25 men between 32 and 89 years old and 22 women between 23 and 89 years old.

Different from the first application, the second dataset is an HAR dataset, which contains accelerometer and gyroscope data collected using a mobile phone [39]. This dataset contains 24 participants ranging from 18 to 46 years old. In contrast to the first dataset, this dataset contains more social context data, including age, gender, height, and weight, where finding suitable grouping manually is challenging.

### B. Data Distribution Analyses

In addition to the age and gender distribution, we further analyzed the distribution across different classes of data in both databases. It is worth noting that it is not uncommon for IoMT applications to have imbalanced data distributions across multiple classes, which may potentially impact the performance evaluation results. The class distributions of the MIT-BIH and HAR datasets are shown in Fig. 2 as the box plot.

Referring to Fig. 2, it is clear that there is a significant imbalance in terms of the number of samples available in each

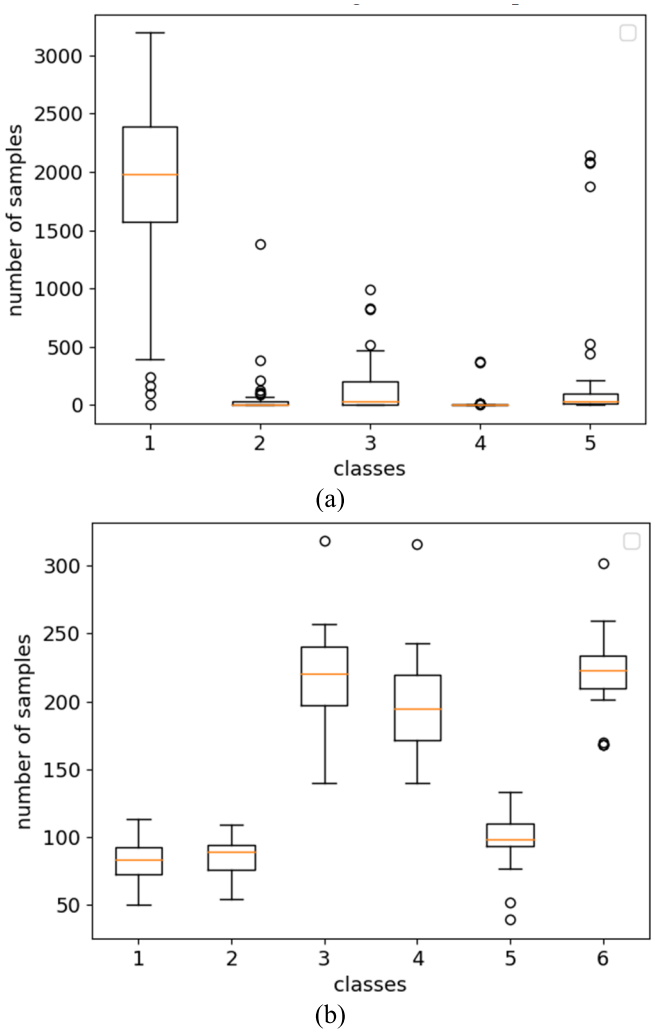



Fig. 2. Class distributions for (a) MIT-BIH dataset and (b) HAR dataset.

class, which is to be expected for typical ECG databases where certain classes (e.g., normal class) will have a lot more data samples than other abnormal classes. Such a different class distribution brings the challenge to ensure fair training and testing and may lead to varying performance results that need to be carefully investigated.

### C. Performance Evaluation of the MIT-BIH ECG Dataset

The purpose of this evaluation is to examine the effectiveness of the proposed framework in processing one type of IoMT applications that has only a small amount of social context data (in this case age and gender), and yet, the class is highly imbalanced. In this type of scenario, participant selection may result in more severe class imbalance, which can lead to an adverse impact on the training results. However, it is also expected that the clustering-based participant selection will enhance the model quality. In this study, the proposed method is compared against a typical horizontal federated learning with no participant selection.

In this evaluation, individual participants conduct their individual training using a typical CNN model with cross-entropy loss used to learn a classifier for different ECG classes.

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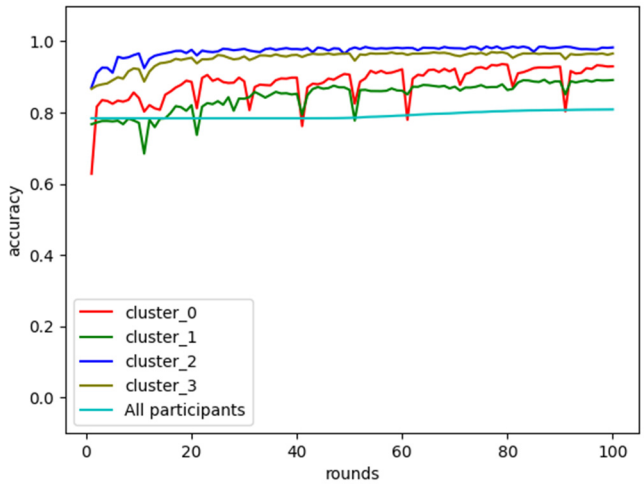



Fig. 3. Federated learning performance evaluation using the MIT-BIH dataset.

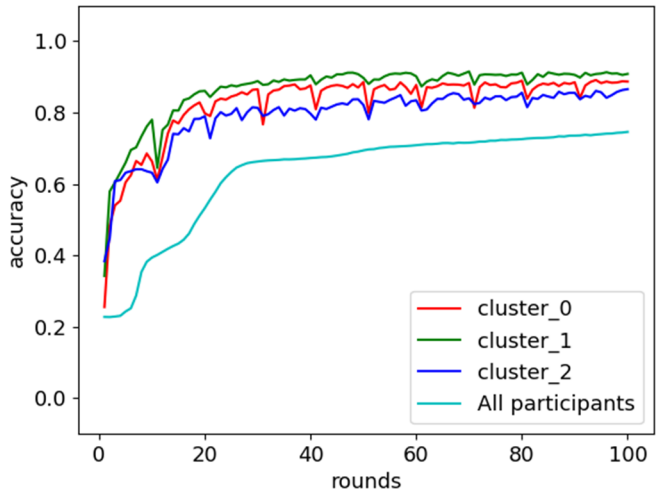



Fig. 4. Federated learning performance evaluation using the HAR dataset.

TABLE III
MIT-BIH FEDERATED LEARNING MODEL ACCURACY WITH AND WITHOUT PARTICIPANT SELECTION

| MIT dataset | Accuracy | No. of participants |
|---|---|---|
| cluster 0 | 0.930 | 16 |
| cluster 1 | 0.892 | 12 |
| cluster 2 | 0.983 | 5 |
| cluster 3 | 0.966 | 5 |
| All participants | 0.808 | 47 |

TABLE IV
HAR FEDERATED LEARNING MODEL ACCURACY WITH AND WITHOUT PARTICIPANT SELECTION

| HAR dataset | Accuracy | No. of participants |
|---|---|---|
| cluster 0 | 0.887 | 8 |
| cluster 1 | 0.908 | 9 |
| cluster 2 | 0.866 | 5 |
| All participants | 0.746 | 24 |

The learning rate is set at 0.001 for the MIT-BIH dataset. All the participants in the MIT-BIH dataset are randomly allocated into two separate groups, representing two separate edge groups. Within each edge group, GMM clustering is performed to separate participants into four different clusters. If a specific cluster has only one participant, the cluster will be removed from the federated learning process due to an insufficient number of participants. Within each cluster group, the participants will perform the first-layer federated learning using FedAvg aggregation among all the participants within the same cluster.

After every tenth iteration, each cluster group will send its aggregated model to the second-layer server to be aggregated with similar clusters from other edge cluster groups. The aggregation of the second-layer federated learning is done using the silhouette coefficient weighting in (3)–(5). The training performance can be shown in Fig. 3 and Table III.

Referring to Fig. 3, it can be observed that at every tenth iteration, there is a fluctuation in the model performance. All four clusters (0–3) show better and faster convergence. These clusters also achieved significantly better accuracy than the baseline model. It is also worth noting that for these clusters, only 34%, 26%, 11%, and 11% of the overall participants are used in the training, and hence, the computation overhead is much less compared to the baseline horizontal federated learning without participant selection.

### *D. Performance Evaluation of the HAR Dataset*

Different from the ECG application, the HAR application dataset has a slightly more balanced class distribution, as shown in Fig. 2(b). However, it also has lesser number of participants and more sophisticated social context data (i.e., age, gender, height, and weight). Overall, the HAR dataset presents a more challenging scenario to the proposed method due to the potential of insufficient data after forming the edge groups with participant selection.

This evaluation follows the same procedures as the previous evaluation where two edge groups are established. Clustering-based participant selection is then performed to form three clusters. Cluster-specific federated learning is then performed at the edge layer. At every tenth iteration, the edge layer models will be aggregated by the second layer (i.e., the cloud server). Since the HAR dataset has less number of participants, only three matching clusters are found in the second-layer model aggregation. The performance of each individual cluster and the baseline model is shown in Fig. 4 and Table IV.

In this application scenario, the formed clusters are also able to achieve better accuracy with a smaller number of participants. In comparison, clusters 0, 1, and 2 used 33%, 38%, and 21% of the total participants to achieve a significantly better performance compared to the baseline method.

Overall, both experimental results demonstrated that: 1) the proposed method is capable of handling different types of IoMT applications; 2) the proposed clustering-based participant selection can effectively select useful participants to achieve more efficient and more accurate federated learning;

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and 3) with effective participant selection, better performance can be achieved with lesser number of participants.

## VI. Conclusion

The modern IoT-based end-edge-cloud infrastructure offers great opportunities to many applications, such industrial automation, smart cities, and remote healthcare. Wearable and mobile devices can constantly sense and collect social, physical, and physiological data to monitor a person's mental and physical health status, forming the so-called IoMT applications. While real-time data can be collected, personal data alone are insufficient to train a good quality model. The cloud-based centralized learning approach also cannot offer efficient and high-quality services. Emerging federated learning is able to address these challenges by providing a more secure way for distributed knowledge sharing. However, it is critical to ensure that suitable participants are involved during the knowledge sharing and aggregation process to avoid an adverse impact on the global model. In this study, a clustered hierarchical federated learning approach is proposed. In our approach, participants are first categorized into multiple learning groups using a GMM clustering algorithm based on their social context information. Individual groups (or clusters) then conduct their own federated learning. Periodically, the models of all learning groups will be shared and aggregated by the cloud server using a unique aggregation function based on the distances between learning groups. The developed framework is evaluated using two publicly available, real-world databases, namely, the MIT-BIH and HAR, which represent different types of IoMT applications. The results demonstrated that the proposed hierarchical federated learning with clustering-based participant selection is able to achieve better accuracy with fewer participants and less amount of data.

As discussed in the experimental evaluation, it is common for IoMT applications to have imbalanced datasets. Different social context information may also have different impacts depending on the targeting IoMT applications. Our future works will further enhance the federated learning process to evaluate the algorithm with different IoMT applications with varying social context data.

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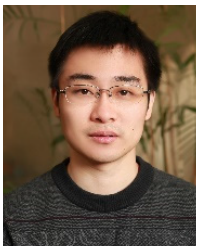
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